

Jiayu Chen

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Education

University of Georgia, Athens, Georgia
Ph.D. in Electric Computer Engineering

Aug 2024 - Present
GPA: 4.0/4.0

Hohai University, Nanjing, China
B.Eng. in Software Engineering

Sep 2020 - May 2024
GPA: 4.80/5.00

Projects

Vital Signs Estimation for Adults and Babies

2024-2025

Focused on accurate vital sign estimation from BCG signals, including heart rate (HR), respiratory rate (RR), inter-beat interval (IBI), and heart rate variability (HRV), while developing strong hands-on expertise with commonly used physiological signals, including PPG, ECG, and respiratory belt signals.

Sleep Apnea Monitoring

2025

Led development of a real-time overnight sleep monitoring system using noisy BCG signals, covering feasibility validation, clinical data collection, signal processing, algorithm development, backend architecture, AI deployment, and FDA regulatory documentation. Delivered continuous snoring detection, apnea event detection, and sleep-wake classification, producing clinically relevant metrics including total sleep time (TST), apnea-hypopnea index (AHI), and sleep apnea severity classification.

- Led feasibility studies and real-world clinical data collection to validate system performance in practical deployment scenarios.
- Designed signal processing and machine learning algorithms for signal denoising, signal quality assessment, occupancy detection, continuous vital sign monitoring, and sleep apnea analysis.
- Built components of the AI inference and deployment pipeline to support real-time continuous monitoring applications.
- Prepared technical documentation and contributed to regulatory submissions and scientific manuscripts, including materials supporting FDA 510(k) premarket notification.

Selected Publications

UbiComp 2025	<i>SelfDenoiser: Self-supervised Seismic Signal Denoiser for Continuous and Contactless Cardiac Monitoring.</i> Jiayu Chen , Yingjian Song, Yida Zhang, Zixuan Zeng, Xiang Zhang, Zaid Pitafi, Zaipeng Xie, Deepak Das, Nishan Dong, Junjie Lu, Xiao Yin, WenZhan Song.
Percom 2026	<i>Attention Feature Fusion with Cluster Contrastive Learning for Snoring and Breath-Holding Detection Using Seismic Sensing.</i> Yingjian Song*, Jiayu Chen *, Yida Zhang, Zixuan Zeng, Zaid Pitafi, Bradley Phillips, Xiang Zhang, Wenzhan Song. (* Equal contribution.)
UbiComp 2026	<i>Contactless Sleep Apnea Detection with Bodyseismography.</i> Yingjian Song*, Jiayu Chen *, Yida Zhang, Zixuan Zeng, Zaid Pitafi, Bradley Phillips, Xiang Zhang, Fei Dou, Wenzhan Song. (* Equal contribution.)
CinC 2026	<i>SeatVitals: Geophone-Based Unobtrusive Vital Sign Monitoring for Office Environments.</i> Yingjian Song*, Jiayu Chen *, Yida Zhang, Zixuan Zeng, Zaid Pitafi, Bradley Phillips, Xiang Zhang, Zaipeng Xie, Qin Lu, Fei Dou, Wenzhan Song. (* Equal contribution.)
Under Review	<i>SeismoApnea: Bodyseismography-Based Overnight Sleep Apnea Monitoring and Apnea-Hypopnea Index Estimation.</i> Jiayu Chen , Zixuan Zeng, Yingjian Song, Jiahui Li, Zhenlin An, Xiang Zhang, Fei Dou, Deepak Das, Bradley Phillips, Wenzhan Song.

Skills

Research	Signal Processing, Time-Series Analysis, Human Sensing, Efficient Deep Learning
Programming	Python, C, SQL, LaTeX
Tools	Linux, InfluxDB, MQTT, Grafana, Git